# ■ Python Challenge

***Instructions:***

• Total Questions: 5

• Use meaningful variable names.

• Write clean and optimized code.

• Do not use external libraries unless mentioned.

• Add comments wherever necessary.

• **40 marks for each question.**

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## Question 1: Rotate a List K Times

**Problem Statement:**
Given a list of integers and an integer K, rotate the list to the right by K positions.

**Example 1**
**Input:**

```
nums = [1, 2, 3, 4, 5]
k = 2
```

**Output:**

```
[4, 5, 1, 2, 3]
```

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## Question 2: Check Palindrome List Using Recursion

**Problem Statement:**
Write a recursive function to determine whether a given list is a palindrome. A list is called a palindrome if it reads the same forward and backward.

**Example 1**
**Input:**

```
nums = [1, 2, 3, 2, 1]
```

**Output:**

```
True
```

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## Question 3: LeetCode Problem

**Problem Statement:**
Go through the link and solve the problem on LeetCode:

```
https://leetcode.com/problems/divide-array-into-equal-pairs/description/
```

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## Question 4: Find the First Non-Repeating Element

**Problem Statement:**
Given a list of integers, find the first element that occurs only once. If no such element exists, return -1.

**Example 1**
**Input:**

```
nums = [4, 5, 1, 2, 0, 4, 1, 2]
```

**Output:**

```
5
```

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## Question 5: Count Frequency of Each Element

**Problem Statement:**
Given a list of integers, count how many times each element appears in the list and store the result in a dictionary.

**Example 1**

**Input:**

```
nums = [1, 2, 2, 3, 3, 3]
```

**Output:**

```
{
    1: 1,
    2: 2,
    3: 3
}
```